Multispectral Image Pan-Sharpening Guided by Component Substitution Model

Huiling Gao, Shutao Li, *Fellow, IEEE*, Jun Li, *Fellow, IEEE*, and Renwei Dian, *Member, IEEE*

***Abstract*— Multispectral image pan-sharpening aims to increase the spatial details of multispectral images by fusing multispectral and panchromatic (PAN) images. Existing component substitution (CS)-based deep learning pan-sharpening is generally regarded as a black box and fails to mine the image interaction relation with physical significance in each step of pan-sharpening, which not only limits the improvement of image resolution but also ignores the physical interpretability of the models. To improve this situation, according to the traditional CS-based detail injection pan-sharpening model, we consider the matrix calculation in each step as the transformation between image pixel values and carry out linear transformations, and therefore the pan-sharpened multispectral image is represented as the sum of two multispectral images. Then given the spatial and spectral heterogeneity, the two summed images are decomposed based on the fact that any real number can be expressed as the product of two real numbers. Ultimately, the multispectral image pan-sharpening model can be constructed as the sum of two Hadamard products. We design a dual-branch network with attention mechanisms that merges the sum and the Hadamard products into a concise formulation. This method not only enhances physical interpretability but also improves spatial resolution. Experiments on five real-world datasets validate that the proposed multispectral image pan-sharpening model can improve performance.**

***Index Terms*— Attention mechanism, Hadamard product, multispectral image pan-sharpening, two-branch network.**

I. Introduction

THE high-resolution multispectral (HRMS) images can enhance the interpretation ability of the image to achieve reliable results. However, the hardware limitations of the imaging system directly lead to the trade-off among spectral resolution, spatial resolution, and signal-to-noise ratio (SNR), which inevitably reduces the applied range of MS images. Therefore, the fusion method both keeping the spectral resolution and improving the spatial resolution becomes an effective method of acquiring an HRMS image. It fuses a low-spatial-resolution (LRMS) image and a high-spatial-resolution panchromatic (PAN) image [1], [2], and the pan-sharpened HRMS image is expected to have the same spectral and spatial resolution as the LRMS and PAN image, respectively, which significantly improves the performance of scene classification and interpretation, target detection, and so on.

Traditional component substitution (CS) refers to substituting the LRMS information with high-spatial-resolution information from a PAN image. Over the past decades, pan-sharpening techniques have emerged [3], [4], [5], [6], and been integrated into a unified detail injection framework [1]. The methods have clear physical significance, but there are many faults, such as spectral mismatch and poor representation ability, which greatly suppress the improvement of performance.

As the strong learning ability of the convolutional neural network (CNN) is exploited [7], numerous excellent models based on CNN have been constructed. The super-resolution CNN (SRCNN) [8] first transforms the image super-resolution problem into learning the mapping relationship between input and output images by an end-to-end CNN. The multispectral image pan-sharpening aims to fuse multispectral image and PAN image to produce a multispectral image with high resolution. Thus, these works [9], [10] only regard image pan-sharpening as learning a mapping relationship. With the emergence of various complex models with high performance, novel network architectures [11], [12], [13], [14], [15] are introduced into it. However, compared to traditional pan-sharpening methods, they introduce less spectral distortion and greatly enhance spatial details but consider little physical interpretability. While pursuing high accuracy using more complex models, the efficiency and application range are greatly reduced. Although more researchers [16], [17], [18], [19] have been interested in interpretable models and the constructed deep learning frameworks follow the primary detail injection [20], [21], [22], [23], they lack exploration in the internal transformation process.

In this work, we explore the traditional modeling process of detail injection and transform the traditional CS technology into the sum of two Hadamard products through linear transformation between pixel values, and then the integrated pan-sharpening model is implemented by a double-branch CNN. Each Hadamard product is expressed as the combination of

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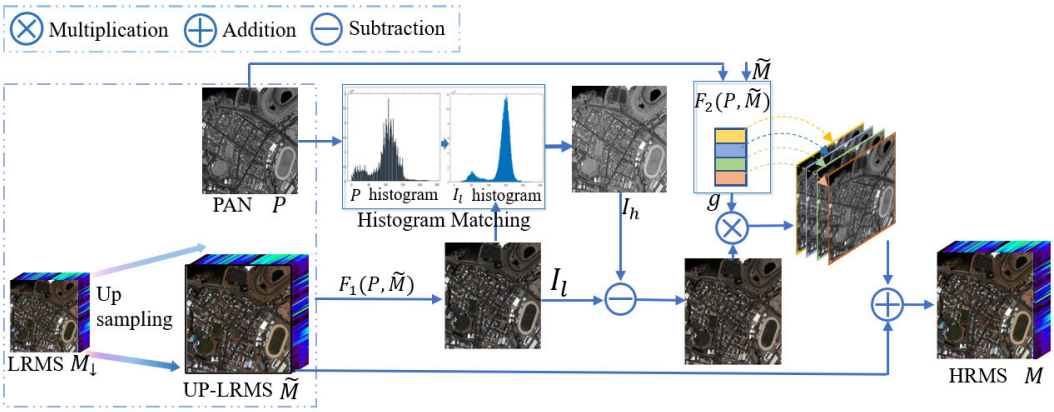


Fig. 1. Generic flowchart for pan-sharpening founded on the CS method.

a pan-sharpened image and its attentions, where the attention mechanisms can better reflect the spatial and spectral differences of images. Therefore, when the traditional CS-based model is transformed using a linear transformation, we pay full attention to the physical interpretability. In addition, the whole multispectral image pan-sharpening model is transformed into a CNN to improve the multispectral image resolution.

II. Related Work

A. Traditional CS Fusion

CS approaches aim at improving the resolution of MS images by replacing certain components in a given domain. First, the upsampled MS image is implemented the space transformation that the spatial component is segregated from the spectral information. To produce less distortion, the histogram matching must be executed to ensure a greater correlation between the PAN image and the chosen component. Next, the specific components of a MS image are performed substitution using the matched PAN image. In the end, the substituted MS image is transformed into the original image space through the inverse transformation [24]. CS is also called projection substitution, representative CS methods include intensity-hue-saturation (IHS) [3], principal component analysis (PCA) [25], Brovey transform (BT) [26], and Gram-Schmidt (GS) [4]. Tu et al. [5] propose a new CS formalization and then it is analyzed in subsequent works. Based on the assumption of linear transformation and single CS, it is shown that the fusion process can be obtained through a proper injection scheme, even though these forward and backward transformations are implicit. The injection scheme can keep higher fidelity and lead to faster implementation.

As shown in Fig. 1, traditional CS approaches can be summed up in the detail injection structure [1]. $\mathbf{P} \in R^{H\times W}$ is a PAN image, $M_\downarrow$ represents a LRMS image, let $\tilde{M} \in \mathrm{R}^{H\times W\times C}$ denote a spatially interpolated LRMS image according to the size of the PAN image, where C is the band number. $\hat{\mathbf{M}} \in \mathrm{R}^{H\times W\times C}$ denotes the pan-sharpened output. The process can be generally expressed as

$$\hat{\mathbf{M}}_k = \tilde{\mathbf{M}}_k + g_k(\mathbf{I}_h - \mathbf{I}_l), \quad k = 1, \ldots, C \tag{1}$$

where $\hat{\mathbf{M}}_k$ and $\tilde{\mathbf{M}}_k$ is the kth band of $\hat{\mathbf{M}}$ and $\tilde{\mathbf{M}}$, respectively, $\mathbf{g} = [g_1, \ldots, g_k, \ldots, g_C]$ is the replacement gain. $\mathbf{I}_h \in R^{H\times W}$ denotes a histogram-matched image with $\mathbf{P}$ and the intensity component $\mathbf{I}_l \in R^{H\times W} = F_1(\tilde{\mathbf{M}})$ obtained by the following equation:

$$F_1(\tilde{\mathbf{M}}) = \sum_{k=1}^{C} w_k \tilde{\mathbf{M}}_k \tag{2}$$

where the weight vector $\mathbf{w} = [w_1, \ldots, w_k, \ldots, w_C]$ represents the degree of spectral overlap of PAN and MS channels and comes from the first row of the forward transform matrix. Based on the basic framework, later scholars improve performance by enhancing the whole process or a certain part. Lu et al. [6] explore the gradient and detail relationship between a PAN image and each band of the MS image. Addesso et al. [27] develop a procedure to estimate both the parameters of the residual distribution and the injection coefficients. Vivone et al. [28] design the slope of the regression line as the coefficients. Restaino et al. [29] apply a nonlinear injection procedure. Xiao et al. [30] estimate adaptive gain coefficients.

B. CNN Based Fusion

Deep learning-based methods have a strong fitting ability and have achieved good performance improvement. Giuseppe et al. [9] apply the idea of mapping to pan-sharpening and propose a CNN pan-sharpening model (PNN). To improve performance, Kim et al. design a residual CNN (ResNet) structure and Wei et al. [10] put forward a deep residual neural network (DRPNN) for pan-sharpening. Jiang et al. [31] improve insufficient spatial enhancement based on ResNet. To overcome the dependence of the model on data, Luo et al. [32] present a novel unsupervised iterative CNN-based pan-sharpening method. Zero-reference GAN (ZeRGAN) [33] introduce multiscale generators and discriminators with nonreference loss functions to implement fusion. The papers [13], [34] construct a transformer-based model for pan-sharpening. In addition, attention mechanisms [35], [36], [37] are widely used in deep learning-based tasks due to high performance. Qu et al. [38] introduce a self-attention mechanism into the pan-sharpening model to further make full use of the rich spectral information of MS images. Guan and Lam [39] adopt a block with spectral and spatial attention mechanisms to fuse the features efficiently. Recently,

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Deng et al. [40] conduct a full review of the advance of deep learning-based methods for the application of pan-sharpening. In multispectral image pan-sharpening, the wavelength range is not equal for an MS image and a PAN image. There are differences in spatial detail distribution and among spectral bands. Therefore, attention is a powerful tool to discriminate spatial details and spectral channel differences.

The physical interpretability of models has attracted extensive attention. It is inspired by the mechanism of the traditional detail-injected framework, Yang et al. [41] incorporate domain-specific knowledge and focus on the preservation of spectral and spatial information. Deng et al. [21] design a deep CNN fusion scheme based on the traditional CS and multiresolution analysis. Feng et al. [42] map a two-step optimization model to several detailed convolutional blocks for detail injection. He et al. [20] design a detail injection-based CNN network (DiCNN). Xiang et al. [43] establish a network employing nonlinear detail learning and detail injection. Tian et al. [19] construct a variational pan-sharpening network with clear physical meanings. Cao et al. [23] integrate sparse coding and an alternative algorithm into the network. Cao et al. [23] replace the intensity component and the injection gain using a CNN.

III. Proposed Model

In this section, we provide pan-sharpening details about our method. According to the detail injection mechanism, as shown in Fig. 1, the histogram equalization is performed on the PAN image $\mathbf{P}$ as shown in the following equation:

$$\begin{aligned} s_d &= T(r_d) = (L-1)\sum_{j=0}^{d} p_r(r_j) \\ &= \frac{L-1}{HW}\sum_{j=0}^{d} n_j, \quad d = 0, \ldots, L-1 \end{aligned} \tag{3}$$

where T is an increasing gray transformation, r_d is the dth gray level of the image $\mathbf{P}$, s_d represents dth gray level of the image $\mathbf{I}_s \in \mathrm{R}^{H\times W}$ after histogram equalization. L means the maximum gray level and p_r denotes the probability density function of gray level of the image $\mathbf{P}$. n_j is the number of pixels with gray value r_j. Then according to the probability density function p_l of gray level of the reference image $\mathbf{I}_l$, we can obtain the transformation function as shown in the following equation:

$$G(z_q) = (L-1)\sum_{i=0}^{q} p_l(z_i) \tag{4}$$

z is the gray level of the image $\mathbf{I}_h$, and q indicates the gray level index of the reference image. We hope that the histogram equalization image has the same gray level as the reference image and $s_d = G(z_q)$. To get pixel value z_q of the image $\mathbf{I}_h$, we perform the transformation $z_q = G^{-1}(s_d)$ on each pixel with gray level.

From the above, the entire histogram matching process is regarded as the transformation of the corresponding pixel values among the input image $\mathbf{P}$, the histogram-equalization image $\mathbf{I}_s$ and the histogram-matching image $\mathbf{I}_h$. No matter how the transformation functions T and G change, we only pay attention to the transformation relationship between pixels. Thus, the mapping between pixel values i_r and i_s in these two images $\mathbf{P}$ and $\mathbf{I}_s$ is shown in the following equation:

$$i_s = \begin{cases} a_{ir} * i_r, & i_r \neq 0 \\ b_{ir}, & i_r = 0 \end{cases} \tag{5}$$

where $a_{ir}, b_{ir}(>= 0) \in \mathrm{R}$ are parameters and * represents pixel product. The relationship of each pixel between two images $\mathbf{I}_s$ and $\mathbf{I}_h$ is shown in the following equation:

$$i_z = a_{is} * i_s + b_{is} \tag{6}$$

where i_z is the pixel value of the image $\mathbf{I}_h$ and a_{is}, b_{is} $(>= 0) \in \mathrm{R}$ indicate transform coefficients. We can get the transformation relationship between pixels of the input image and matched image as shown in the following equation:

$$\begin{aligned} i_z &= a_{is} * (a_{ir} * i_r + b_{ir}) + b_{is} \\ &= a_{is} * a_{ir} * i_r + a_{is} * b_{ir} + b_{is} \\ &= a * i_r + b \end{aligned} \tag{7}$$

where the parameter b can be calculated by the following equation:

$$b = \begin{cases} 0, & G^{-1}(s_0) = z_0 = 0 \\ b_l i_l, & G^{-1}(s_0) > 0 \end{cases} \tag{8}$$

where i_l and b_l are pixel value and its coefficient of the template image $\mathbf{I}_l$. The relationship of the image $\mathbf{I}_h$ matched by histogram and the input image $\mathbf{P}$ is expressed in the following equation:

$$\mathbf{I}_h = \mathbf{A} \odot \mathbf{P} + \mathbf{B}_l \odot \mathbf{I}_l \tag{9}$$

where $\mathbf{A} \in \mathrm{R}^{H\times W}$ and $\mathbf{B}_l \in \mathrm{R}^{H\times W}$ are coefficient matrices composed of coefficients a and b_l of each pixel, respectively. The symbol "$\odot$" represents Hadamard product.

For $\mathbf{I}_l$, similar to the above transformation process, we only focus on the corresponding relationship between pixel values of the two images $\mathbf{I}_l$ and $\tilde{\mathbf{M}}_k$, therefore, the image $\mathbf{I}_l$ can be formulated as the linear transformation of any band $\tilde{\boldsymbol{M}}_k k = 0, \ldots, C$. The fixed value image $\mathbf{I}_l$ is also obtained by corresponding band $\tilde{\boldsymbol{M}}_k$

$$\mathbf{I}_l = \mathbf{U}_k \odot \tilde{\mathbf{M}}_k + \mathbf{V}_k \tag{10}$$

where $\mathbf{U}_k \in \mathrm{R}^{H\times W}$ is a coefficient matrix and $\mathbf{V}_k \in \mathrm{R}^{H\times W}$ is a constant matrix with the same dimension as $\tilde{\mathbf{M}}_k$. If gray values of all pixels in the image $\tilde{\mathbf{M}}_k$ are either greater than 0 or equal to 0, then $\mathbf{I}_l$ is obtained through the corresponding pixel transformation between images, such as the first part on the right side of (10). When gray values of some pixels in the image $\tilde{\mathbf{M}}_k$ are equal to 0 and gray values of some corresponding pixels in the image $\mathbf{I}_l$ are greater than 0,

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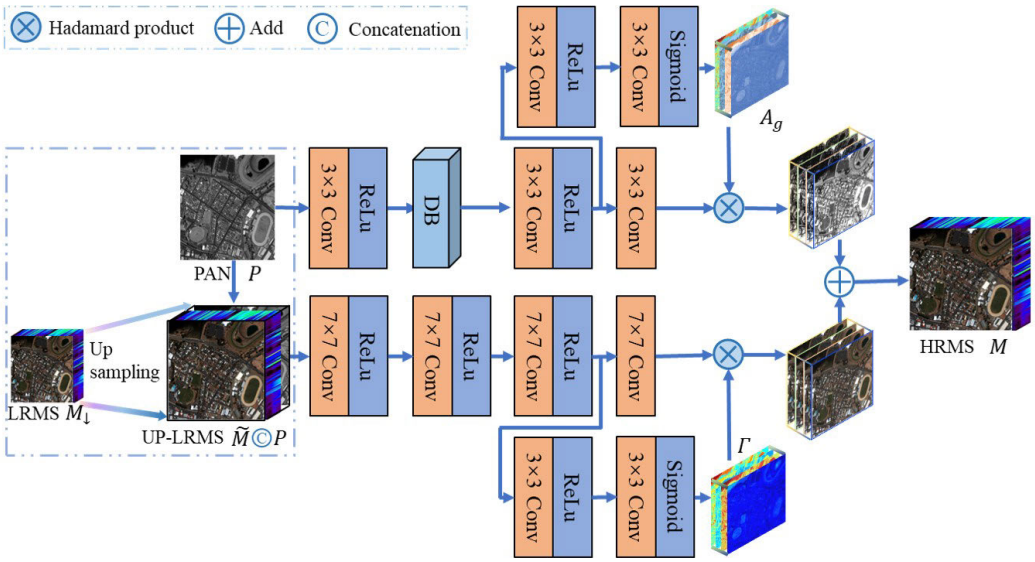


Fig. 2. Flowgraph of our method.

we substitute (9) and (10) into (1)

$$\begin{aligned}\hat{\mathbf{M}}_k &= \tilde{\mathbf{M}}_k + g_k(\mathbf{A}\odot\mathbf{P} + \mathbf{B}_l\odot\mathbf{I}_l - \mathbf{I}_l)\\ &= \tilde{\mathbf{M}}_k + g_k(\mathbf{A}\odot\mathbf{P} + (\mathbf{B}_l - \mathbf{E})\odot\mathbf{I}_l)\\ &= \tilde{\mathbf{M}}_k + g_k\left(\mathbf{A}\odot\mathbf{P} + \mathbf{B}_l^I\odot\mathbf{I}_l\right)\\ &= \tilde{\mathbf{M}}_k + g_k\left(\mathbf{A}\odot\mathbf{P} + \mathbf{B}_l^I\odot\left(\mathbf{U}_k\odot\tilde{\mathbf{M}}_k + \mathbf{V}_k\right)\right)\\ &= \left(\mathbf{E} + g_k\mathbf{B}_l^I\odot\mathbf{U}_k\right)\odot\tilde{\mathbf{M}}_k + g_k\mathbf{B}_l^I\odot\mathbf{V}_k + g_k\mathbf{A}\odot\mathbf{P}\\ &= \mathbf{M}_\mathbf{V} + g_k\mathbf{A}\odot\mathbf{P}\\ &= \mathbf{H}_k\odot\widehat{\mathbf{M}}_k + \mathbf{A}_g\odot\tilde{\mathbf{P}} \end{aligned} \tag{11}$$

where $\mathbf{E}$ is a unit matrix, $\mathbf{B}_l^I = \mathbf{B}_l - \mathbf{E}$, $\tilde{\mathbf{P}} \in R^{H\times W\times C}$ and $\mathbf{M}_\mathbf{V} = (\mathbf{E} + g_k\mathbf{B}_l^I\odot\mathbf{U}_k)\odot\tilde{\mathbf{M}}_k + g_k\mathbf{B}_l^I\odot\mathbf{V}_k$ are intermediate variables. $\mathbf{M}_\mathbf{V} \in R^{H\times W\times C}$ is decomposed, and the values $h_k \in [0, 1]$ of the decomposed matrix $\mathbf{H}_k$ are assured by the function Sigmoid(.). Similarly, the value range of matrix $\mathbf{A}_{gk}$ is 0–1. Consequently, the estimated HRMS image is $\hat{\mathbf{M}}$

$$\hat{\mathbf{M}} = \mathbf{\Gamma}\odot\widehat{\mathbf{M}} + \mathbf{A}_g\odot\tilde{\mathbf{P}} \tag{12}$$

where $\mathbf{A}_g \in R^{H\times W\times C}$, $\mathbf{\Gamma} \in R^{H\times W\times C}$.

The variables of the whole fusion process in (12) can be learned by CNN. As shown in Fig. 2, according to (12), the estimated HRMS image is equal to the sum of two parts and the process can be executed by a double-branch CNN. We expect the first part on the right side of (12) to keep the spectral information as much as possible and enhance the spatial information, and the concatenated images $\tilde{M}©P$ are used to learn $\hat{\mathbf{M}}$ as shown the branch at the bottom in Fig. 2. Another branch is used for learning the last part on the right side of (12), it not only makes up for the details and slightly poor spectral information that cannot be extracted in the first mapping process, but also greatly improves the convergence speed at the beginning of training. Multispectral images contain spatial and spectral characteristics, and each band has smooth regions and high-frequency details. Because it is difficult to completely recover the high-frequency details, we expect to give them different weights of attention. In the spectral dimension, each spatial pixel corresponds to a spectral curve, and different objects have different spectral curves. Therefore, we expect to use the attention mechanisms to implement differential attention to each curve. Considering the differences in spectral bands and spatial details of multispectral images, the attention mechanisms $\mathbf{\Gamma}$ and $\mathbf{A}_g$ are utilized. The designed pixel-by-pixel attention can not only reflect the importance of bands but also give importance evaluation to the detailed information in each band. For network architecture, we prefer a simple network architecture. The top branch contains three convolutional layers and a dense block (DB) [14] with two convolutional layers. The DB can interact with different level features and a single input $\mathbf{P}$ need not consider the issue of spectrum aliasing. The bottom branch only employs four convolutional layers.

The whole fusion process is optimized through minimizing the loss function $\mathcal{L}(\theta)$

$$\mathcal{L}(\theta) = \left\|\mathbf{\Gamma}\odot\widehat{\mathbf{M}} - \mathbf{M}\right\|_F^2 + \left\|\hat{\mathbf{M}} - \mathbf{M}\right\|_F^2 \tag{13}$$

where $\mathbf{M}$ is the ground truth image HRMS, θ is the CNN parameters to be optimized.

IV. Experimental Results

A. Experiment Setting

1) Experimental Database: We simulate experiments on five real-world remote sensing images from four sensors. The datasets contain three small and two large datasets with four and eight bands.

WorldView-2 (WV2 Washington) dataset is taken over an urban area in Washington D.C. using the WV2 sensor, which provides an eight-band MS image, and a PAN image with a wavelength range from 450 to 800 nm. The spatial resolutions are 1.85 and 0.46 m. The sensor's radiometric resolution is 11 bits. In the reduced-resolution experiments, a 2400 × 2400 MS region and a 256 × 256 MS area are utilized to train and test. In full-resolution test experiments, the MS scene with 220 × 220 pixels is selected.

IKONOS (IK Hobart) dataset is from the IK sensor, which shows an urban and harbor area of Hobart in Australia. It supports a four-band MS image and a PAN image in a wavelength

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range of 450 to 900 nm. The spatial resolution of MS and PAN are 4 and 1 m, and there is a 11-bit radiometric resolution. Two MS areas including sizes of 2400 × 2400 pixels and 256 × 256 pixels are used to train and test in the reduced-resolution experiments. A 130 × 130 MS region is selected for full-resolution test experiments.

Quickbird (QB Sundarbans) dataset from QB sensor with 11 bits provides a forest area of Sundarbans in India. It contains a PAN image and a four-band MS image and their resolutions are 0.6 and 2.4 m, respectively. Two MS areas with the sizes of 2400 × 2400 and 256 × 256 pixels are selected to train and test in the reduced-resolution experiments. A 220 × 220 MS region is for full-resolution test experiments.

WorldView-3 (WV3) dataset is from a public dataset [21] and it is acquired by the WV3 sensor. The spatial resolutions are 0.3 m for the PAN image and 1.2 m for the MS bands with eight bands. The radiometric resolution is 11 bits. The dataset contains 9714 image pairs with sizes of 64 × 64 and 16 × 16 for training, of which 30% is used for verifying. There are 20 image pairs with sizes of 256 × 256 and 64 × 64 for testing in the reduced-resolution. In the full-resolution test experiments, the dataset provides 20 image pairs with sizes of 512 × 512 and 128 × 128. In addition, to verify the generalization ability of the model, we also conduct performance evaluation on the public WV2 dataset [44] with different scenes, where images in two different scenes are used for the reduced-resolution experiments and the sizes of image pairs are 256 × 256 and 64 × 64. Furthermore, 16 image pairs with sizes of 512 × 512 and 128 × 128 are employed to evaluate full-resolution performance.

GF2 public dataset [21] is collected by GF-2 satellite sensor with a radiometric resolution of 10 bits and it is obtained over the city of Beijing. The spatial resolutions are 4 and 1 m for the MS and PAN images, respectively. There are 19 809 image pairs with sizes of 64 × 64 and 16 × 16 for training and two test datasets containing 20 image pairs with sizes of 256 × 256 and 64 × 64 are used for reduced-resolution, while 20 image pairs with sizes of 512 × 512 and 128 × 128 are provided to the full-resolution test experiments. QB dataset [21] is acquired over the city of Indianapolis cutting it into two parts based on QB satellite sensor. The radiometric resolution is 10 bits and spatial resolutions are 2.4 and 0.6 m for the MS and PAN images. The image size is the same as that of GF2 dataset and QB dataset contains 17 139 image pairs for training. Twenty image pairs are used for reduced-resolution test. The MS images have four bands and therefore we merge the two datasets for training, of which 30% is for verifying.

In the experiment, we follow Wald's protocol [45]. In reduced-resolution experiments, the obtained PAN and MS images are degraded by a Gaussian filter and subsampled with a factor of 4 in the spatial domain, and then the downsampled MS images are up-interpolated by four times. The processed PAN and MS images are used as inputs to the network.

2) Quantitative Metrics: Based on fair performance evaluation, we use recognized and open metrics including $Q2^n$ [46], spatial correlation coecient (SCC) [47], spectral angle mapper (SAM) [48], root mean square error (RMSE), and relative dimensionless global error in synthesis (ERGAS) [49]. Among them, SCC is used to evaluate the spatial information of the image, the SAM and $Q2^n$ indexes are used to evaluate the spectral information of the image, and ERGAS and RMSE can evaluate the overall quality of the image. Furthermore, the performance is evaluated at full resolution and we employ the quality without reference (QNR), D_λ and D_s indexes [1], where the QNR is a jointly spectral and spatial quality index, D_λ and D_s indexes are a spectral distortion index and a spatial distortion index, respectively.

TABLE I
PARAMETERS OF OUR NETWORK STRUCTURE

		Layer	Filter	Stride	Output Size
Input	PAN				32*32*1
	MSI				32*32*4
Feature extraction	PAN	Conv1+Relu	3*3/64	1	32*32*64
		DB			32*32*96
		Conv2+Relu	3*3/64	1	32*32*64
		Conv3	3*3/64	1	32*32*4
Spatial attention		Conv4+Relu	3*3/64	1	32*32*64
		Conv5+Sigmoid	3*3/4	1	32*32*4
		Hadamard product			32*32*4
Feature extraction	MSI+PAN	Conv6+Relu	7*7/64	1	32*32*64
		Conv7+Relu	7*7/64	1	32*32*64
		Conv8+Relu	7*7/64	1	32*32*64
		Conv9	7*7/4	1	32*32*4
Spatial attention		Conv10+Relu	3*3/64	1	32*32*64
		Conv11+Sigmoid	3*3/4	1	32*32*4
		Hadamard product			32*32*4
Restoration		Add			32*32*4

3) Compared Methods: For our comparison, we choose classical models based on traditional CS methods including band dependent spatial detail (BDSD) [50], GS adaptive (GSA) [4], partial replacement adaptive component substitution (PRACS) [51], and context-aware details injection fidelity (CDIF) [30], besides, several deep learning-based methods are also involved, such as component substitution network (CSNET) [21], PNN [9], DRPNN [10], pan-sharpening network (PanNet) [41], DiCNN [20], and spatial and spectral extraction (SSE) [52].

4) Experimental Details: The deep learning framework Keras and PyTorch are used for deep learning-based pan-sharpening methods, traditional methods are accomplished and performance indexes are computed using MATLAB R2018b in the Windows 10 operating system.

The parameters of the network structure are given in Table I, such as on QB dataset. We employ Adam [53] to minimize the objective function in (13). In training, the patch sizes of small and large datasets are 32 × 32 and 64 × 64, the stride is adjusted to 8, the batch size is 32, and initial learning rate is 5e-4.

Fig. 3 shows the training loss plots of MS image pan-sharpening methods based on the shallow networks on WV2, IK, and QB datasets. To make observation and comparison more clearly, we enlarge the loss image at 30–50 epoch. Obviously, our method exhibits lower training error.

B. Reduced Resolution Assessment

We evaluate the performance of our method on three small datasets and two large datasets containing single-sensor datasets and mixed-sensor datasets.

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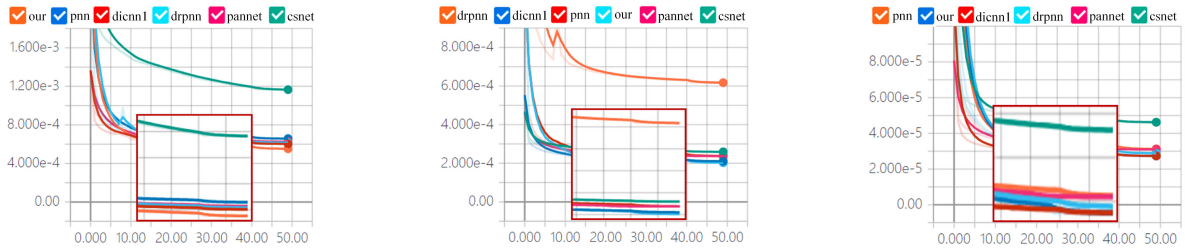


Fig. 3. Training loss plots of the methods compared on WV2, IK, and QB Datasets. (a) WV2. (b) IK. (c) QB.

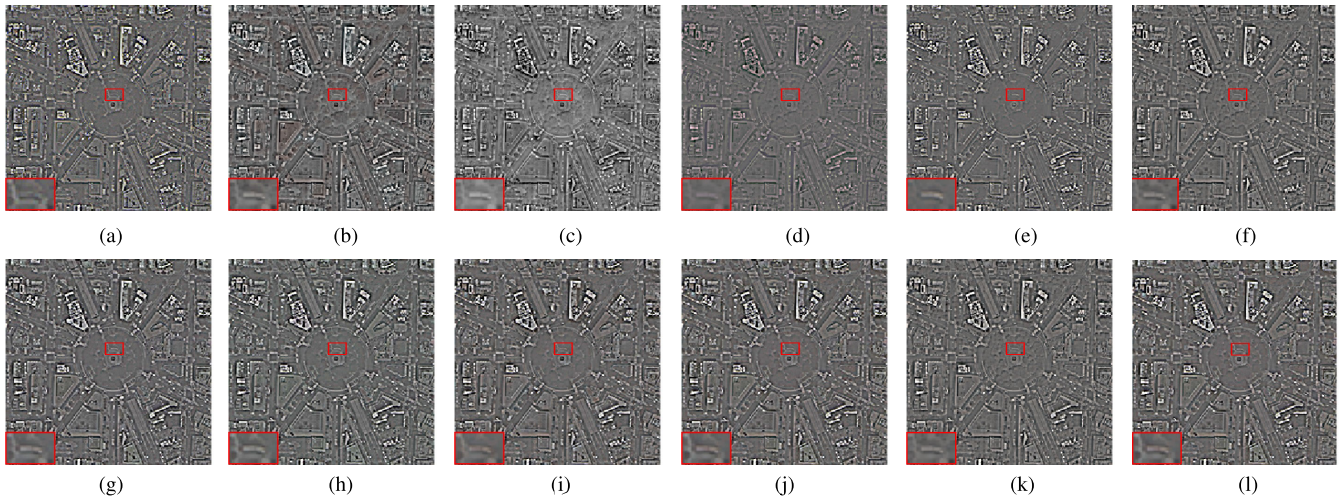


Fig. 4. False-color learned detail images (composed of bands 3, 2, and 1) of the pan-sharpened results. (a) Ground-truth. (b) BDSD. (c) GSA. (d) PRACS. (e) CSNET. (f) CDIF. (g) PNN. (h) DRPNN. (i) PanNet. (j) DiCNN. (k) SSE. (l) Ours.

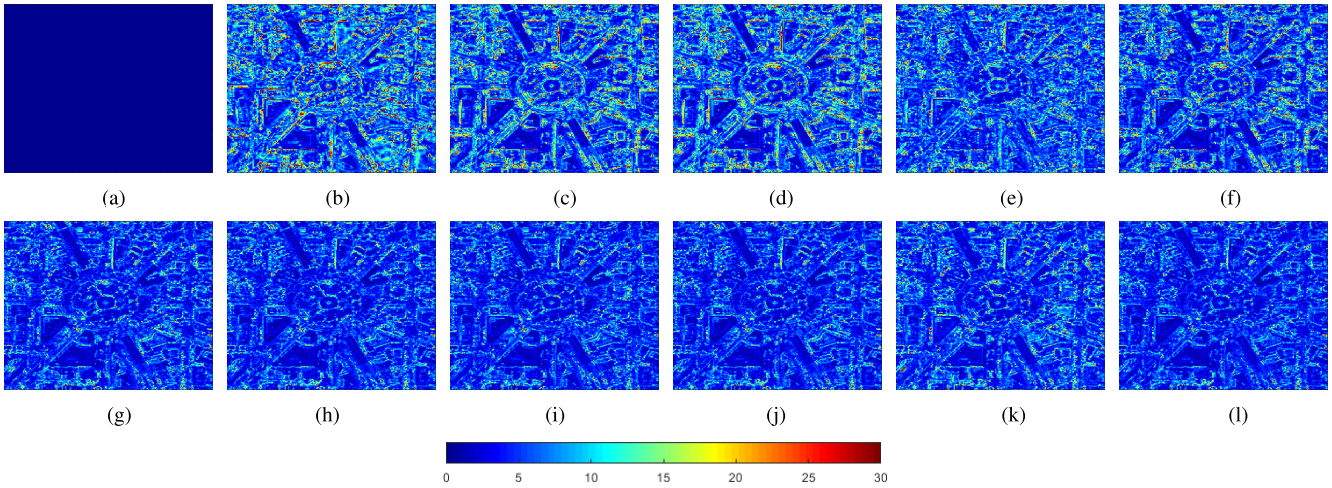


Fig. 5. SAM pan-sharpened images on WV2 dataset. (a) Ground-truth. (b) BDSD. (c) GSA. (d) PRACS. (e) CSNET. (f) CDIF. (g) PNN. (h) DRPNN. (i) PanNet. (j) DiCNN. (k) SSE. (l) Ours.

1) Experiments on WorldView-2 Washington Dataset: As shown in Table II, our method gets higher scores than the CS-based methods. For CSNET, part of the traditional component replacement method is replaced by CNN, which transfers the error based on the conventional feature transformation to the later network. Compared with the end-to-end CNN methods, the scores of our method's quality indexes are closer to the reference values, which benefits from the principle of our method following the traditional CS and reasonable neural network architecture that can fit each parameter derived from different modules. Our model is also compared with the deep model SSE, and the experimental results show that a moderate network depth will show better performance.

The learned detail images are shown in Fig. 4, where the detail image of the ground truth is formed by the residual image of the ground truth and the interpolated LRMS. Similarly, other detail images are from the pan-sharpened HRMS and interpolated LRMS image. It can be seen clearly that the learned spatial details contain spatial distortion. In Fig. 4, the red box in the lower right corner shows the enlarged content of the small red box in the middle of the figure. As can be seen from the ground-truth details, the curved short lines in the red box are very blurred, by contrast, the ones of the

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TABLE II
PERFORMANCE COMPARISON OF THE DIFFERENT PAN-SHARPENING METHODS ON A SUBSCENE OF WV2 DATASET

	$Q2^n$	SAM	ERGAS	SCC	RMSE
Refrence	1	0	0	1	0
BDSD	0.9208	8.1188	4.5325	0.8992	81.5294
GSA	0.9195	7.5830	4.3501	0.8973	78.7366
PRACS	0.8708	7.7315	5.2647	0.8650	95.7267
CSNET	0.9293	6.5580	4.0560	0.9145	72.8043
CDIF	0.9321	6.6747	3.9718	0.9099	71.4382
PNN	0.9652	4.9647	2.8087	0.9603	50.2287
DRPNN	0.9663	4.8817	2.7598	0.9622	49.2659
PanNet	0.9656	4.8898	2.7920	0.9633	49.7942
DiCNN	0.9668	4.8441	2.7386	0.9629	48.9282
SSE	0.9552	5.6256	3.2126	0.9486	57.2824
Our	**0.9682**	**4.7250**	**2.6871**	**0.9656**	**47.8978**

TABLE III
REDUCED-RESOLUTION PERFORMANCE COMPARISON OF THE DIFFERENT PAN-SHARPENING METHODS ON IK DATASET

	$Q2^n$	SAM	ERGAS	SCC	RMSE
Refrence	1	0	0	1	0
BDSD	0.8385	5.4006	4.2047	0.7584	71.2164
GSA	0.8379	5.1063	4.1468	0.7550	70.4690
PRACS	0.8163	5.1182	4.2097	0.7646	71.1451
CSNET	0.8285	5.5055	4.4087	0.8844	73.6936
CDIF	0.8565	4.6603	3.8369	0.7892	65.2911
PNN	0.9500	3.3657	2.3416	0.9367	39.8910
DRPNN	0.9558	3.1743	2.1986	0.9465	37.4507
PanNet	0.9485	3.3827	2.3655	0.9381	40.4151
DiCNN	0.9562	3.1578	2.1887	0.9469	37.3251
SSE	0.9277	4.1928	2.9305	0.9039	49.7117
Our	**0.9601**	**3.0316**	**2.0981**	**0.9527**	**35.7660**

other detail images are clear. The reason is that these methods overemphasize the details in the learning process. As shown above, the distortion of spatial information is inevitable in the process of learning the mapping relationship, but how to effectively reduce the distortion is necessary. For detailed images of the compared methods, our result is the closest to the curved short line in the ground-truth details. Fig. 5 shows pan-sharpened SAM images. Overall, the high-resolution images restored by deep learning methods contain more detailed information than the traditional detail injection methods.

2) Experiments on IKONOS Hobart Dataset: Table III displays the calculated values of performance evaluation indexes for different methods. It is evident that the values of the CNN-based methods are closer to the reference values. That is based on the fact that the traditional CS methods focus on substitution and cannot fit the relationship between the images, while CNN can learn the mapping relationship between image pairs. Therefore, the traditional methods lack the representation ability, which limits the improvement of performance.

Fig. 6 exhibits learned details from the residual image between pan-sharpened images and interpolated MS images. The images in the red boxes in the top right are enlarged in the left bottom and can clearly show the difference among test mages. In terms of edge in the images, the results of PanNet and our method are the most similar to the ground truth. The deep model SSE lacks shallow details, so its performance is poor. Fig. 7 shows the pan-sharpened SAM results. As observed, the SAM images by BDSD, GSA, PRACS, and CDIF exhibit more inadequate spectral information reflected

TABLE IV
REDUCED-RESOLUTION PERFORMANCE COMPARISON OF THE DIFFERENT PAN-SHARPENING METHODS A SUBSCENE OF QB DATASET

	$Q2^n$	SAM	ERGAS	SCC	RMSE
Refrence	1	0	0	1	0
BDSD	0.8268	3.2706	2.3068	0.8477	23.3475
GSA	0.8210	3.0921	2.2317	0.8528	22.5668
PRACS	0.7884	3.2001	2.3330	0.8413	23.0036
CSNET	0.8960	2.2947	1.5108	0.9448	14.9320
CDIF	0.8587	2.6146	1.8810	0.8842	18.9716
PNN	0.9298	1.7739	1.2440	0.9604	12.3102
DRPNN	0.9257	1.8089	1.2449	0.9605	12.3072
PanNet	0.9350	1.6999	1.2093	0.9638	12.0094
DiCNN	0.9359	1.7044	1.1990	0.9631	11.8851
SSE	0.9162	2.0003	1.3729	0.9546	13.5460
Our	**0.9427**	**1.6008**	**1.1235**	**0.9692**	**11.1401**

in the image color, especially in dense building areas. While CNN-based methods other than CSNET recover both spectral and spatial information due to the stronger ability of feature representation. These models have better spectral fidelity and recover more spatial details. CSNET uses CNN to replace some parts of CS, and its learning ability is limited.

3) Experiments on QB Sundarbans Dataset: There are the reduced-resolution assessment indexes listed in Table IV. Our values are closer to the reference values than other methods, and spectral and spatial information can be recovered well. DiCNN uses the detail injection method to maximize the spatial detail information while maintaining spectral information, effectively preserving spectral fidelity. DRPNN learns the details by a residual network and finally makes use of the convolutional layer to transform the spectral dimension. Our method is derived from the traditional CS method, and an end-to-end network structure is designed to make each process have a certain physical significance. According to the experimental results, compared with the traditional methods, the performance of our method is greatly improved. Compared with other CNN models, it also maintains certain advantages in spatial resolution.

To clearly compare the performance differences of the methods, we show the learned detail images in Fig. 8. Although the edge restored by all methods is fuzzy, PanNet realizes clearer edges than other models. In addition, the learned details are abundant for our method compared to other methods. Fig. 9 represents SAM images of the pan-sharpened results. It can be seen from the figure that our SAM image is closer to the ground-truth SAM image.

4) Performance Curves on the Three Datasets: As can be seen from Fig. 10, our strategy based on the linear transformation is effective.

5) Experiments on WV3 Dataset: Considering the model's generalization ability, we conduct performance evaluation on the large public dataset WV3 and calculate the average performance values. It can be seen from Table V, methods based on deep learning present good performance. By contrast, our method shows advantages on test sets with multiple scenes and most performance indexes are higher than the compared methods. Moreover, we also apply the model to the benchmark dataset WV2 with different scenes. It can be seen from the

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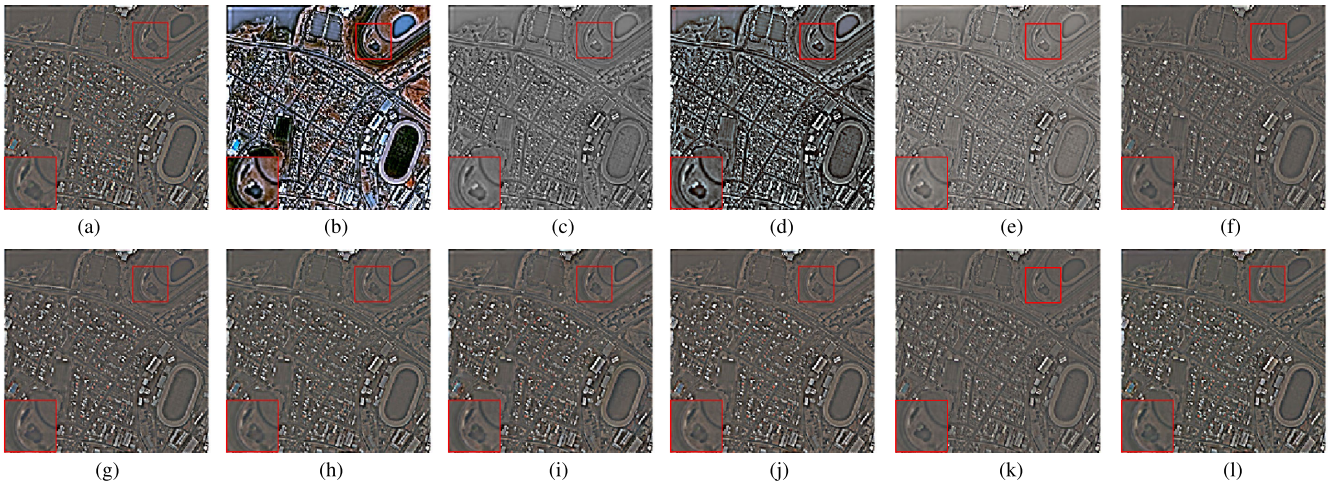


Fig. 6. False-color images (composed of bands 3, 2, and 1) of pan-sharpened detail images on IK dataset. (a) Ground-truth. (b) BDSD. (c) GSA. (d) PRACS. (e) CSNET. (f) CDIF. (g) PNN. (h) DRPNN. (i) PanNet. (j) DiCNN. (k) SSE. (l) Ours.

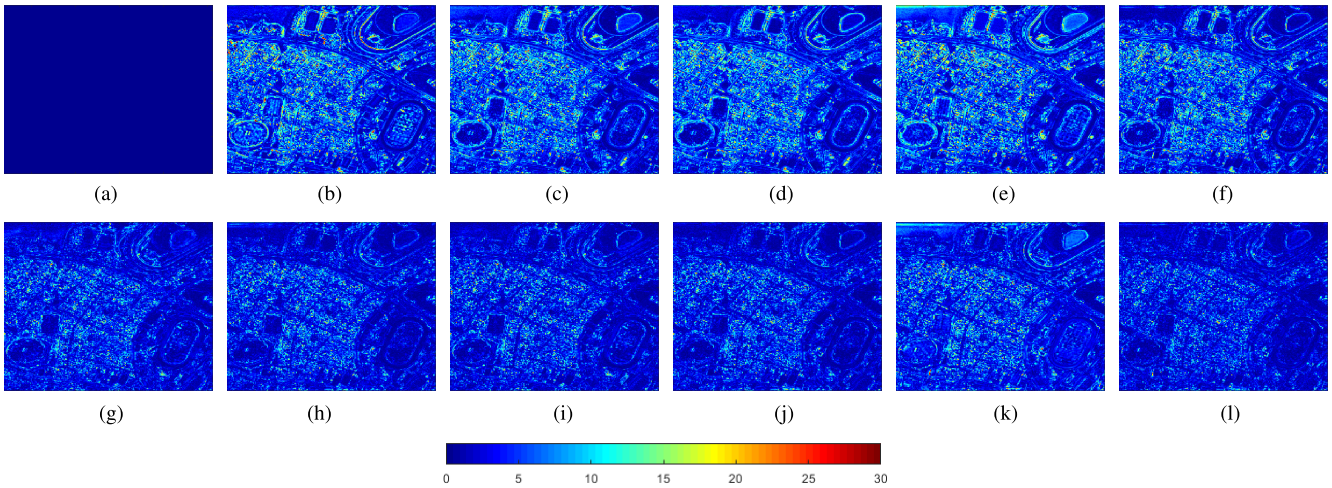


Fig. 7. SAM images of pan-sharpened results on IK dataset. (a) Ground-truth. (b) BDSD. (c) GSA. (d) PRACS. (e) CSNET. (f) CDIF. (g) PNN. (h) DRPNN. (i) PanNet. (j) DiCNN. (k) SSE. (l) Ours.

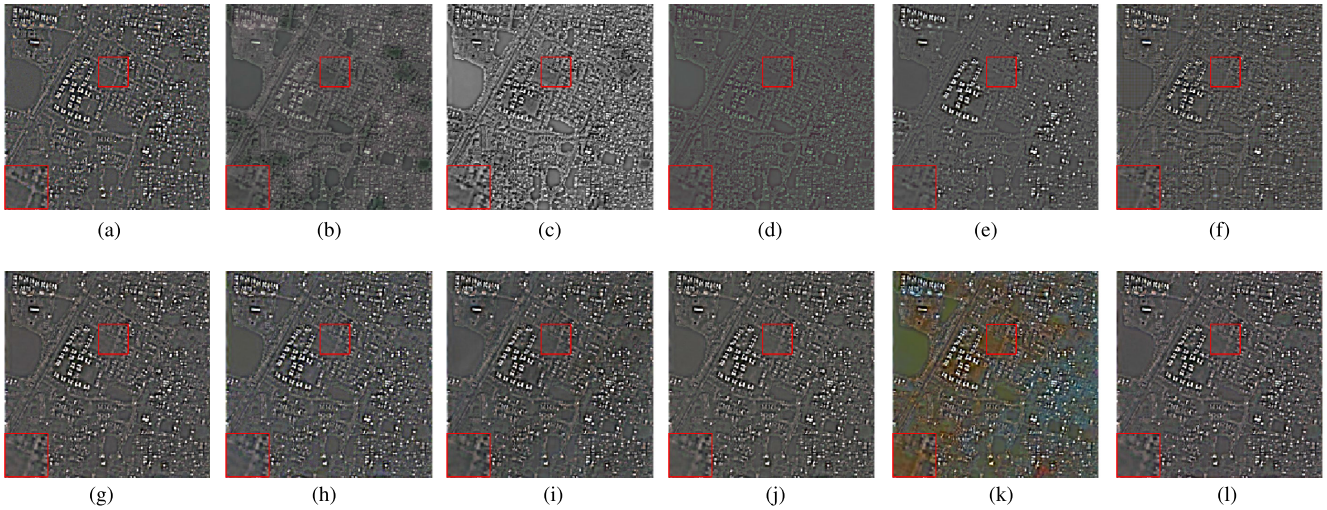


Fig. 8. False-color images (composed of bands 3, 2, and 1) of detail images on QB dataset. (a) Ground-truth. (b) BDSD. (c) GSA. (d) PRACS. (e) CSNET. (f) CFID. (g) PNN. (h) DRPNN. (i) PanNet. (j) DiCNN. (k) SSE. (l) Ours.

objective index values in Table VI that our model can be applied to different scenes and has good generalization ability.

6) Experiments on GF2-QB Dataset: To further evaluate the model's generalization ability, we carry out performance calculation on the large public dataset GF2 and QB (GF2-QB). As shown in Table VII, we provide the average performance index values and we can clearly see that our method shows high values. Similarly, we also calculate the performance indexes on the public benchmark QB. Although the training data is composed of images from different types of sensors

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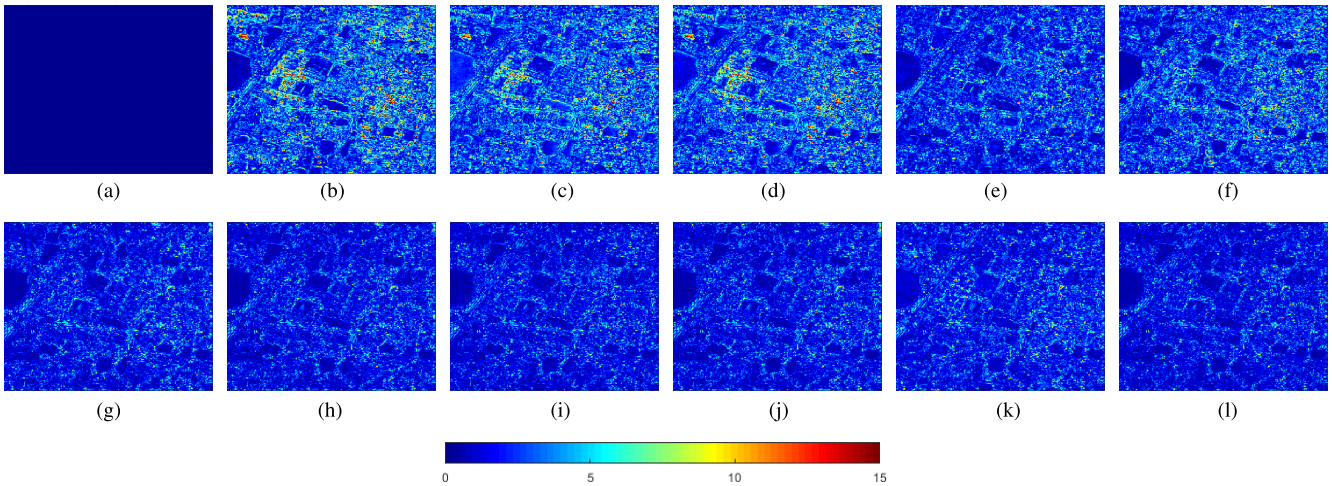


Fig. 9. SAM images of pan-sharpened images on QB dataset. (a) Ground-truth. (b) BDSD. (c) GSA. (d) PRACS. (e) CSNET. (f) CDIF. (g) PNN. (h) DRPNN. (i) PanNet. (j) DiCNN. (k) SSE. (l) Ours.

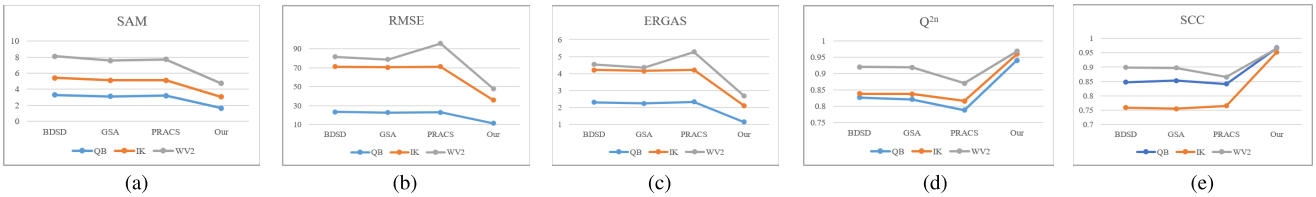


Fig. 10. Reduced-resolution performance curves of the different pan-sharpening methods on WV2, IK, and QB datasets. (a) SAM. (b) RMSE. (c) ERGAS. (d) $Q2^n$. (e) SCC.

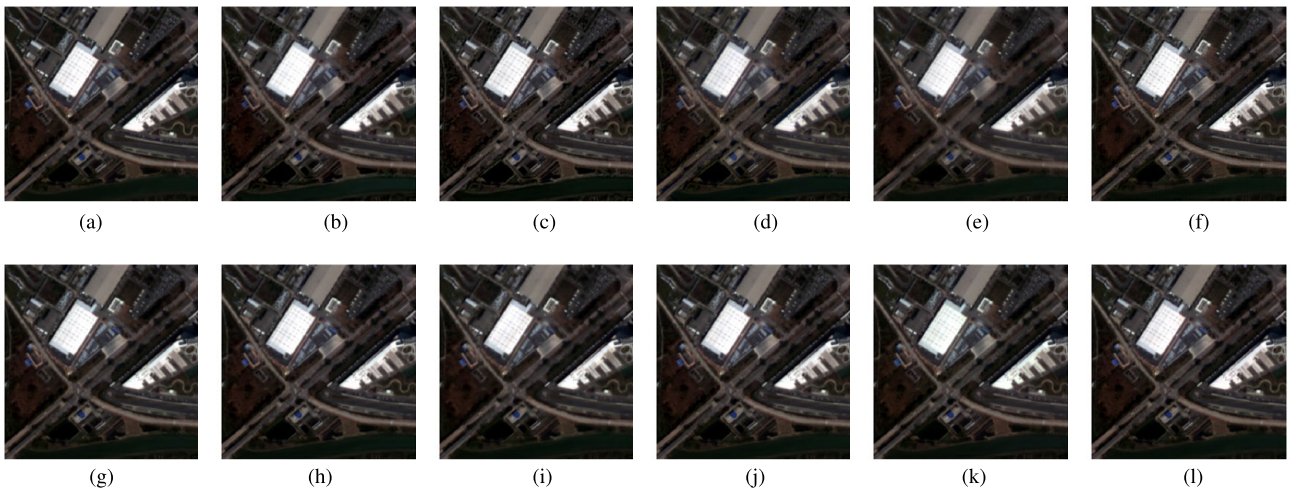


Fig. 11. False-color learned images (composed of bands 3, 2, and 1) of the pan-sharpened results. (a) Ground-truth. (b) BDSD. (c) GSA. (d) PRACS. (e) CSNET. (f) CDIF. (g) PNN. (h) DRPNN. (i) PanNet. (j) DiCNN. (k) SSE. (l) Our.

GF2 and QB, our model can still obtain high-performance fusion results, as illustrated in Table VIII. Fig. 11 shows the false-color fusion images and most of the deep learning methods have achieved good fusion results. They improve resolution without noticeable spatial distortion and our consequences are very similar to the ground-truth. Fig. 12 are pan-sharpened SAM images on GF2 dataset, and our results are the closest to the ground-truth. Especially the quadrilateral and its edge in the upper left corner, we can clearly see that they have the minimum spectral distortion.

C. Full Resolution Assessment

In this section, we evaluate the performance of the proposed method at the original (full) scale. We display and compare fusion results based on the GF2 dataset with four bands and the WV2 dataset with eight bands, where the trained models are the GF2-QB model and the WV3 model, respectively. According to He et al. [20], these evaluation indexes without reference (QNR), D_λ and D_s, they are not perfect, and we can employ them as relative reference indexes to evaluate the fusion ability of the model.

1) Performance on GF2 Datasets: As shown in Table IX, we offer the calculated average nonreference index values on the GF2 dataset. These results show that the fusion ability of the methods based on deep learning is still better than the traditional pan-sharpening technologies. Our method has achieved relatively good results and it shows low spectral distortion. For spatial distortion, our method is similar to most

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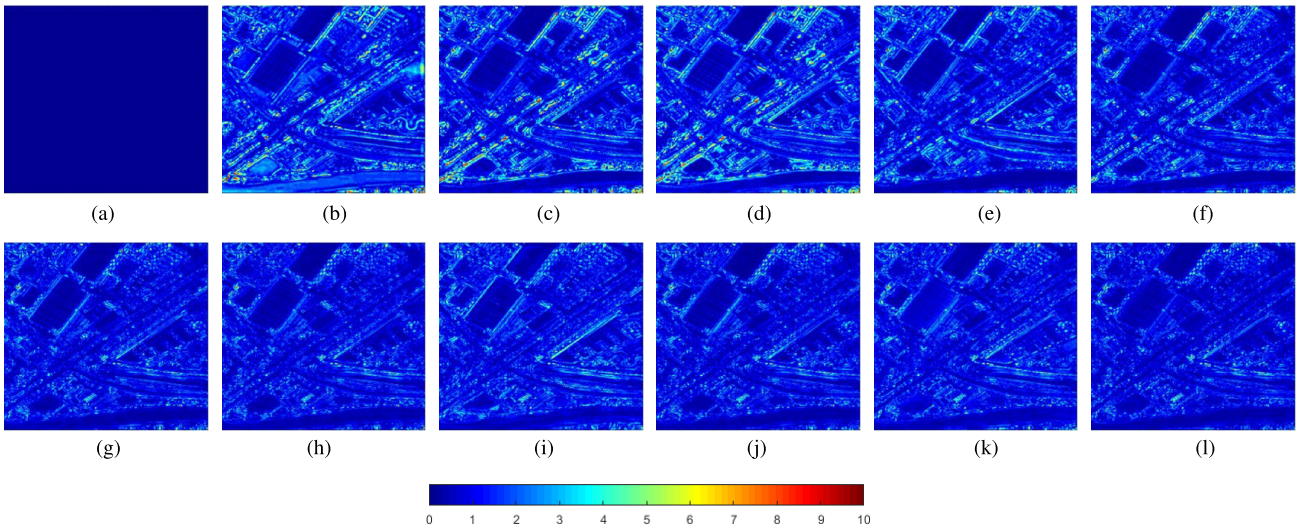


Fig. 12. SAM pan-sharpened images on GF2 dataset. (a) Ground-truth. (b) BDSD. (c) GSA. (d) PRACS. (e) CSNET. (f) CDIF. (g) PNN. (h) DRPNN. (i) PanNet. (j) DiCNN. (k) SSE. (l) Ours.

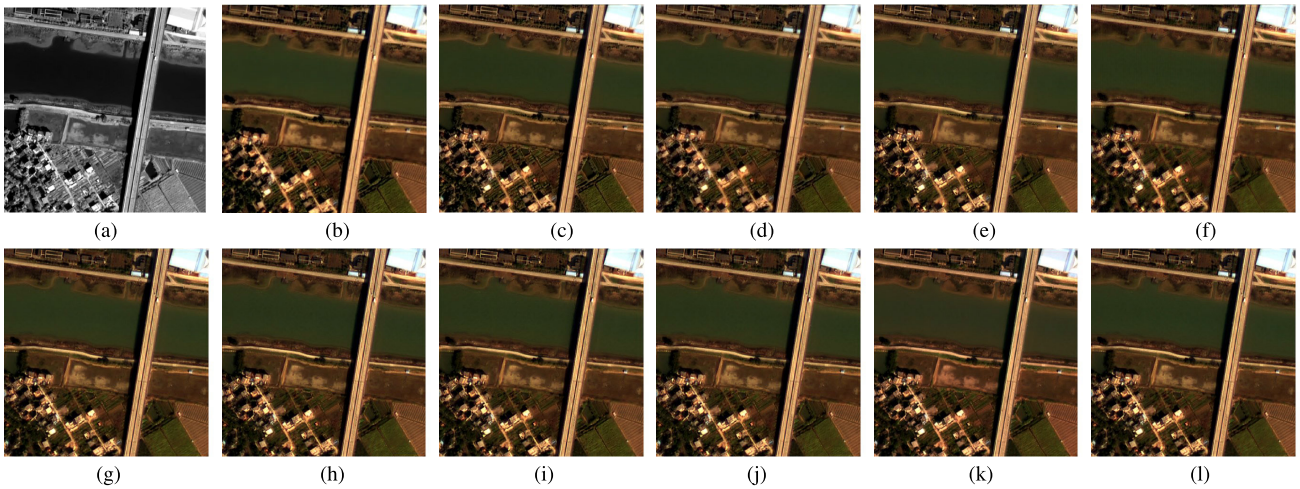


Fig. 13. Nonreference pan-sharpened images on GF2 dataset. (a) Ground-truth. (b) BDSD. (c) GSA. (d) PRACS. (e) CSNET. (f) CDIF. (g) PNN. (h) DRPNN. (i) PanNet. (j) DiCNN. (k) SSE. (l) Ours.

TABLE V

PERFORMANCE COMPARISON OF THE DIFFERENT PAN-SHARPENING METHODS ON THE TESTSET OF WV3 DATASET

	Q2n	SAM	EGRAS	SCC	RMSE
Refrence	1	0	0	1	0
BDSD	0.7780	6.0627	5.7081	0.8301	73.5634
GSA	0.8224	5.3023	4.7188	0.8571	59.6568
PRACS	0.7854	5.5005	5.1956	0.8246	65.5364
CSNET	0.8719	4.1165	3.1899	0.9269	41.2547
CDIF	0.8238	4.6634	4.3859	0.8772	55.3655
PNN	0.8790	3.8443	2.8267	0.9461	36.1735
DRPNN	0.8750	3.8298	2.8280	0.9462	36.1304
PanNet	0.8680	4.0259	3.0040	0.9444	38.4537
DiCNN	**0.8961**	3.6483	2.6774	0.9506	34.5275
SSE	0.8733	4.2058	3.2767	0.9325	41.4896
Our	0.8921	**3.6022**	**2.6103**	**0.9574**	**33.4845**

TABLE VI

PERFORMANCE COMPARISON OF THE DIFFERENT PAN-SHARPENING METHODS ON WV2 DATASET WITH DIFFERENT SCENES

	Q2n	SAM	EGRAS	SCC	RMSE
Refrence	1	0	0	1	0
BDSD	0.8144	5.6110	4.6054	0.8234	60.5613
GSA	0.6707	6.0191	5.6940	0.7090	76.1541
PRACS	0.7742	5.9995	4.8526	0.7781	64.0027
CSNET	0.8914	7.9444	5.1978	0.8958	67.2970
CDIF	0.8875	8.1507	4.9790	0.8928	67.1210
PNN	0.8863	8.2236	5.2304	0.8956	68.1606
DRPNN	0.8879	8.0325	5.1253	0.9007	66.8933
PanNet	0.8697	8.7716	5.7151	0.8665	74.1721
DiCNN	**0.8938**	7.6678	5.0194	0.9054	65.2622
SSE	0.8622	9.4755	6.0875	0.8734	78.4030
Our	0.8921	**7.5644**	**4.9839**	**0.9114**	**64.6859**

methods. The pan-sharpened images without reference are displayed in Fig. 13. The technologies based on deep learning can greatly improve the spatial resolution of images, where DiCNN and our method recover the most details. However, we can see that DiCNN has spatial distortion from the edge of the bridge. Therefore, our method also achieves good fusion results at full resolution.

2) Performance on WV2 Datasets: The average nonreference index values on the WV2 dataset are calculated. As shown in Table IX, these results show that deep learning-based technology can extract rich features and achieve accurate fusion. Our method performs well in spectral and spatial restoration, especially in spectral distortion.

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TABLE VII

PERFORMANCE COMPARISON OF THE DIFFERENT PAN-SHARPENING METHODS ON QB DATASET BASED ON GF2-QB MODEL

	Q2n	SAM	EGRAS	SCC	RMSE
Refrence	1	0	0	1	0
BDSD	0.8176	8.2056	7.8933	0.7927	57.1651
GSA	0.8242	8.1462	7.6970	0.7912	55.5756
PRACS	0.7859	8.2462	8.4219	0.7845	60.9783
CSNET	0.8910	6.1996	5.3738	0.9270	38.4236
CDIF	0.8479	7.0761	7.3226	0.8076	53.4740
PNN	0.9062	5.5106	4.9298	0.9409	35.3266
DRPNN	0.9128	5.3328	4.5728	0.9507	32.7330
PanNet	0.8780	5.9779	6.1024	0.9275	43.8021
DiCNN	0.9155	5.2390	4.5145	0.9517	32.3013
SSE	0.8917	5.5290	5.4593	0.9363	39.2081
Our	**0.9238**	**4.8894**	**4.1056**	**0.9618**	**29.3596**

TABLE VIII

PERFORMANCE COMPARISON OF THE DIFFERENT PAN-SHARPENING METHODS ON GF2 DATASET BASED ON GF2-QB MODEL

	Q2n	SAM	EGRAS	SCC	RMSE
Refrence	1	0	0	1	0
BDSD	0.8985	1.6860	1.5659	0.9290	17.3424
GSA	0.8793	1.7177	1.7262	0.9104	19.2499
PRACS	0.8999	1.6883	1.6245	0.9187	17.8628
CSNET	0.9250	1.3288	1.3942	0.9396	15.7700
CDIF	0.9274	1.3183	1.3278	0.9383	14.5248
PNN	0.9468	1.1211	1.1767	0.9536	13.1058
DRPNN	0.9543	1.0807	1.0657	0.9601	11.8563
PanNet	0.9377	1.2490	1.3096	0.9506	14.5786
DiCNN	0.9549	1.0657	1.0756	0.9605	11.9593
SSE	0.9506	1.2254	1.1493	0.9581	12.7534
Our	**0.9629**	**0.9962**	**0.9112**	**0.9701**	**10.1038**

TABLE IX

PERFORMANCE COMPARISON OF THE DIFFERENT PAN-SHARPENING METHODS ON GF2 AND WV2 DATASETS AT FULL RESOLUTION

	GF2			WV2		
	QNR	D_λ	D_s	QNR	D_λ	D_s
Refrence	1	0	0	1	0	0
BDSD	0.8739	0.0565	0.0745	0.8393	0.1051	0.0620
GSA	0.7680	0.1169	0.1313	0.7980	0.0635	0.1481
PRACS	0.8664	0.0648	0.0741	0.8586	0.0241	0.1202
CSNET	0.9160	0.0348	0.0512	0.8714	0.0391	0.0932
PNN	0.9351	0.0380	0.0281	0.8963	0.0262	0.0796
DRPNN	0.9497	0.0336	**0.0174**	0.9066	0.0265	0.0687
PanNet	0.9290	0.0439	0.0286	0.9022	0.0247	0.0751
DiCNN	0.9434	0.0360	0.0214	0.9079	0.0350	**0.0591**
SSE	0.9096	0.0476	0.0451	0.8087	0.0638	0.1369
Our	**0.9575**	**0.0199**	0.0231	**0.9088**	**0.0241**	0.0687

TABLE X

PERFORMANCE OF WITH OR WITHOUT ATTENTION OF OUR METHOD

Datasets	Methods	$Q2^n$	SAM	EGRAS	SCC	RMSE
WV2	Our_noatten	0.9671	4.8910	2.7520	0.9633	49.0872
	Our	**0.9682**	**4.7250**	**2.6871**	**0.9656**	**47.8978**
IK	Our_noatten	0.9565	3.1205	2.1875	0.9474	37.2559
	Our	**0.9601**	**3.0316**	**2.0981**	**0.9527**	**35.7660**
QB	Our_noatten	0.9401	1.6314	1.1483	0.9672	11.3966
	Our	**0.9427**	**1.6008**	**1.1235**	**0.9692**	**11.1401**

D. Analysis of Attention Mechanism

This part mainly verifies the effectiveness of the attention mechanisms transformed by Hadamard products. According to the experimental results in Table X, the attention contributes to the acquisition of details and the recovery of the spectrum. It can be seen from Fig. 14 that the two attention mechanism branches can recover more spatial details and reduce spectral distortion.

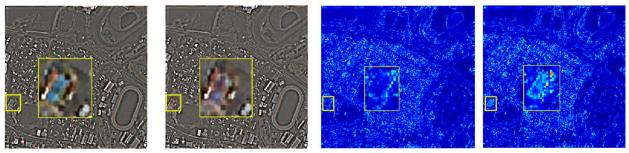


Fig. 14. Reduced-resolution pan-sharpened results with and without attention on IK dataset, where the first two images are detail images, and the other are SAM images.

TABLE XI

SIZE OF TRAINABLE PARAMETERS OF DIFFERENT MODELS

Models	CSNET	DRPNN	PNN	PanNet	DiCNN	SSE
Model size(M)	74.54	227.72	78.54	74.82	223.69	460.92

TABLE XII

SIZE OF TRAINABLE PARAMETERS OF OUR MODEL UNDER DIFFERENT CONDITIONS

Models	Our(7)	Our(7-noDB)	Our(3)	Our(3-noDB)
Model size(M)	557.27	539.27	214.77	196.73

TABLE XIII

PERFORMANCE INDEX VALUES OF OUR METHOD UNDER A SIZE OF 3×3 CONVOLUTIONAL KERNELS ON GF2 AND IK DATASETS

	Q2n	SAM	EGRAS	SCC	RMSE
GF2	0.9558	1.0633	1.0542	0.9625	11.7289
IK	0.9564	3.1517	2.2015	0.9459	37.6153

E. Model Size

Tables XI and XII show the size of trainable parameters of different models and our model under different conditions. In Table XII, the words in brackets represent the convolutional kernel sizes 7×7 and 3×3 in the main branch, and noDB represents that there is no DB module. In all the compared methods, the proposed method lacks model size advantage, mainly due to the kernel size of 7×7. Since we choose a shallow model with a simple network structure, a larger convolutional kernel is conducive to learning more sufficient features as shown in Table XIII.

V. CONCLUSION

A fusion method is presented for the MS and PAN images, founded on this study's traditional CS model. The internal feature transformation in pan-sharpening is converted to linear transformation between pixels, and then we transform the whole pan-sharpening process into the weighted sum of two images. Finally, we design a two-branch CNN to fit the transformed pan-sharpening, in which the attention mechanisms are used to replace the weighting coefficients. The method not only fully considers the physical interpretability by paying close attention to internal conversion processes, but also better improves the resolution of the image by weighted CNN considering the inhomogeneity of spatial details. Extensive experimental results on five real-world public benchmark datasets demonstrate the effectiveness of our method.

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